Do Now · retrieval

Lesson 115 · The Stages of Compilation

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What tool converts assembly mnemonics into machine instructions?

Q2

Why must a binary search operate on ordered data?

Q3

An OR gate has inputs 0 and 1. What is the output?

Q4

What is the effect of enqueue(X), enqueue(Y), dequeue() on an initially empty FIFO queue?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Lexical analysis — Characters become tokens such as identifiers, numbers and operators.

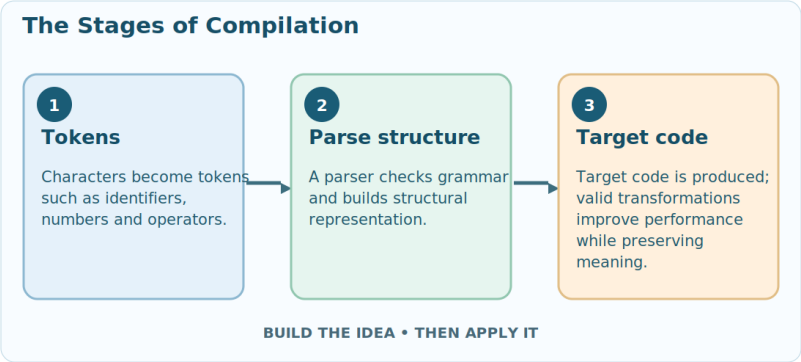
Core idea 2

Syntax analysis — A parser checks grammar and builds structural representation.

Core idea 3

Code generation and optimisation — Target code is produced; valid transformations improve performance while preserving meaning.

Key visual · complete concept



New Learning · applying the ideas

Vocabulary · use the correct technical terms

lexer — Recognises tokens from character stream.

parser — Checks tokens against grammar.

code generation — Production of target instructions.

optimisation — Meaning-preserving transformation to improve an objective.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

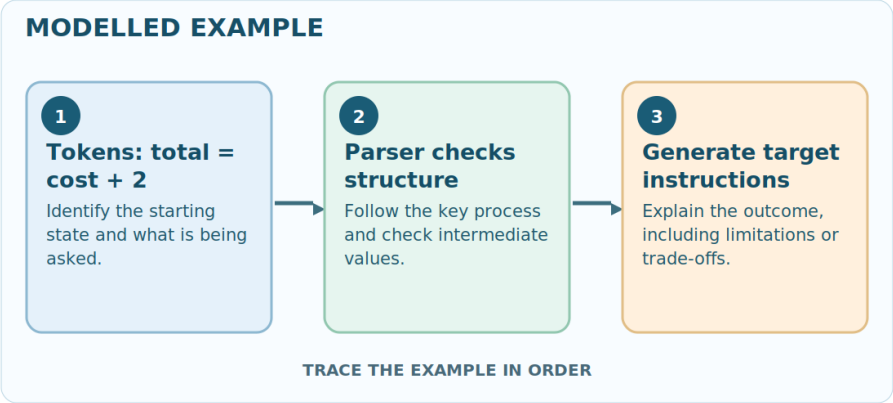
Problem

Explain the stages for the expression 'total = cost + 2' in a high-level source file.

Follow the worked method

1. Model answer — Lexer identifies identifier, assignment, identifier, plus, integer. Parser checks grammatical structure. The compiler creates suitable target instructions; optimisation might fold constants only where semantics permit. — Check: link each claim to the scenario and use precise OCR terminology.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Describe lexical analysis, syntax analysis and code generation. [6]

Question 2

For 'score = score + 1', name five token categories or concrete tokens. [5]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate whether an optimiser may replace 'x = a + 0' with 'x = a', considering language semantics. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A syntax analyser is responsible primarily for reading characters into tokens.

Use key vocabulary from memory

1. Explain lexer in your own words.

2. Explain parser in your own words.

3. Explain code generation in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.